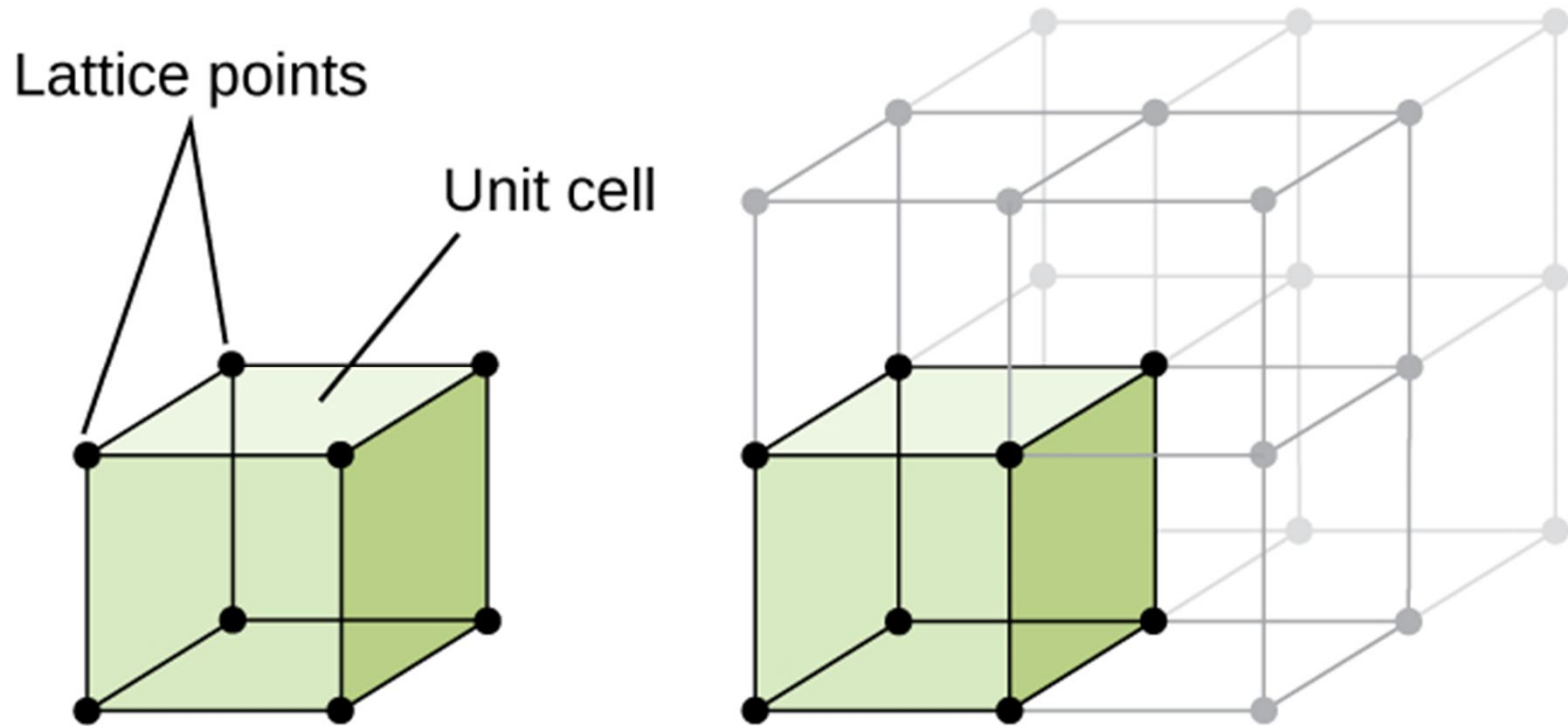


L2: The Crystal Structure of Solids



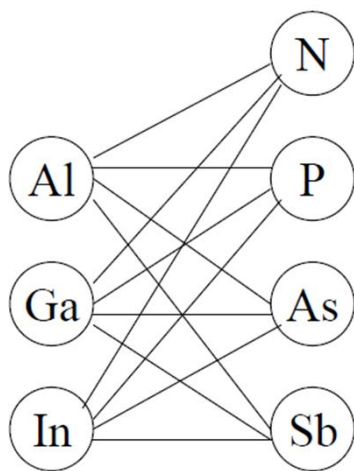
Dr. Shuming Chen

Department of Electrical and Electronic Engineering, SUSTech

Semiconductor materials

Group III	Group IV	Group V
B		
Al	C	P
Ga	Si	As
In	Ge	Sb

A portion of the periodic table



III-V semiconductor

- possible elements for making semiconductor materials

- Si most well known semiconductor. C and Ge can also be semiconductors.

- Other possibilities combinations of III and V group. Eg. GaAs

Elemental and Compound Semiconductors

Elemental: Si, Ge...

Binary: GaAs, GaN, ZnO...

Ternary: $\text{Al}_x\text{Ga}_{1-x}\text{As}$, $\text{In}_x\text{Ga}_{1-x}\text{As}$

Quaternary: $\text{In}_{1-x}\text{Ga}_x\text{As}_{1-y}\text{P}_y$

Types of Solids

- Some solids are *amorphous* with no periodic structure. Atoms or molecules are randomly oriented.
- Crystalline solids are *periodic*.
 - Atoms or molecules are arranged in an orderly and repeating fashion.
 - They have definite shapes with clearly defined faces.
- Polycrystalline solids have small regions of crystalline material.

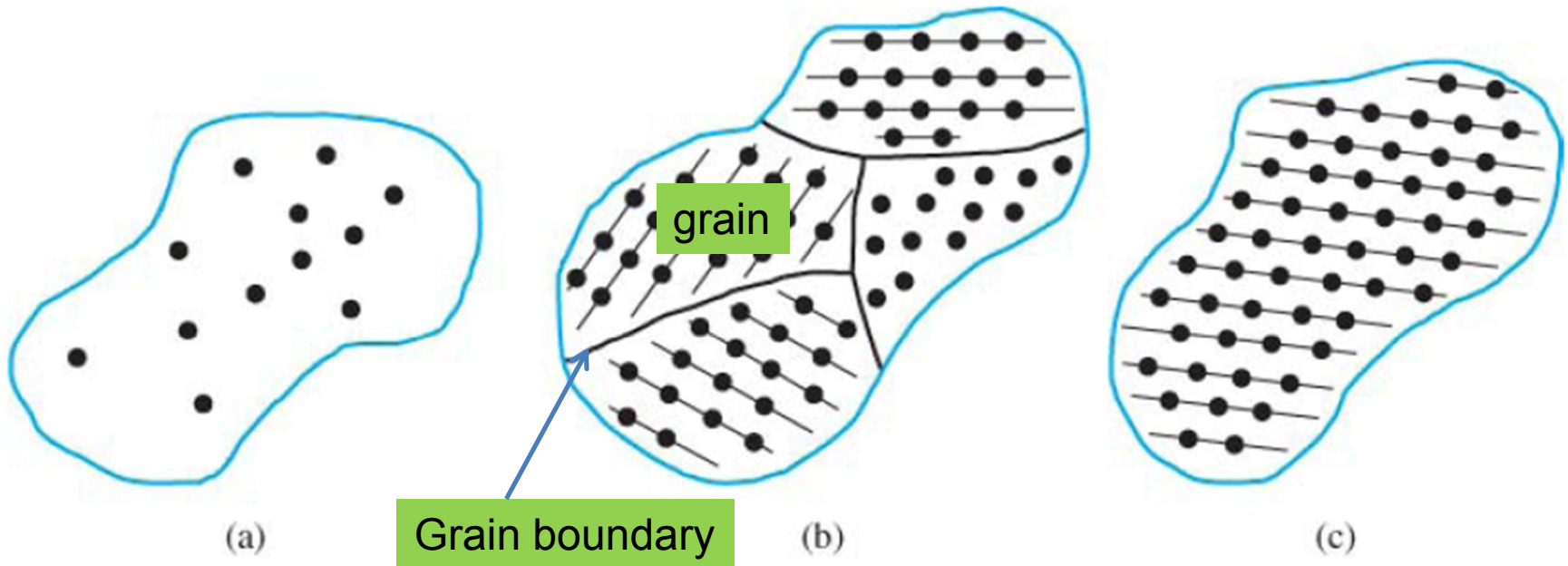


Figure 1.1 | Schematics of three general types of crystals: (a) amorphous, (b) polycrystalline, (c) single.

Amorphous Si

Advantages

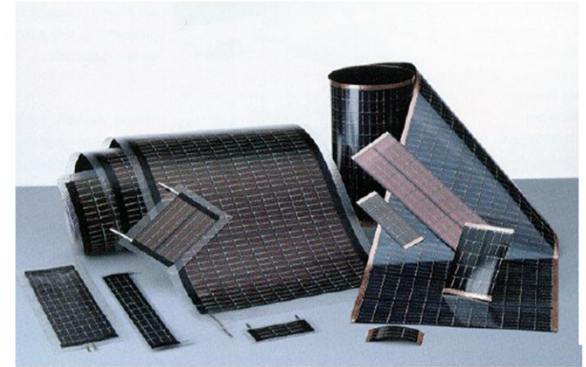
- Low temperature deposition ($\sim 250^{\circ}\text{C}$)
- Large Area deposition possible at low cost

Disadvantages

- Low conductivity
- Cannot be used for High speed circuits

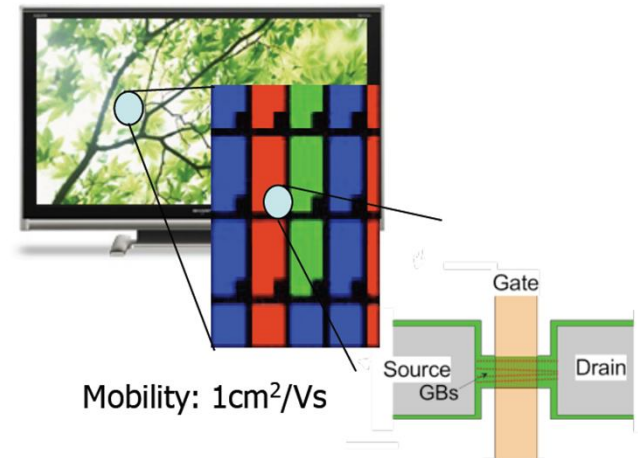
Applications: Solar cells and display driving devices

Solar cells



Efficiency: 6-10%

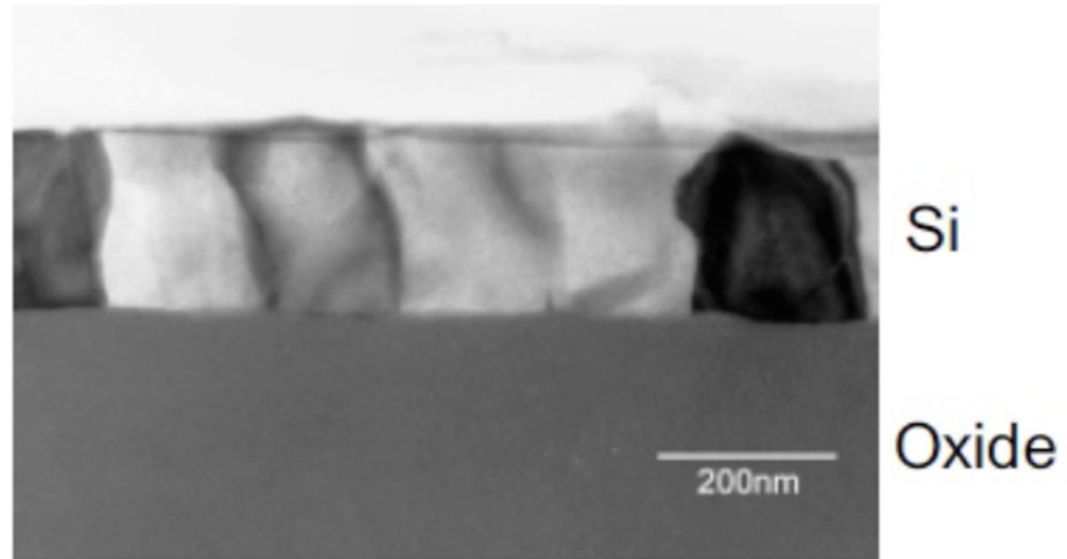
Thin Film Transistors for LCDs



Polycrystalline Si

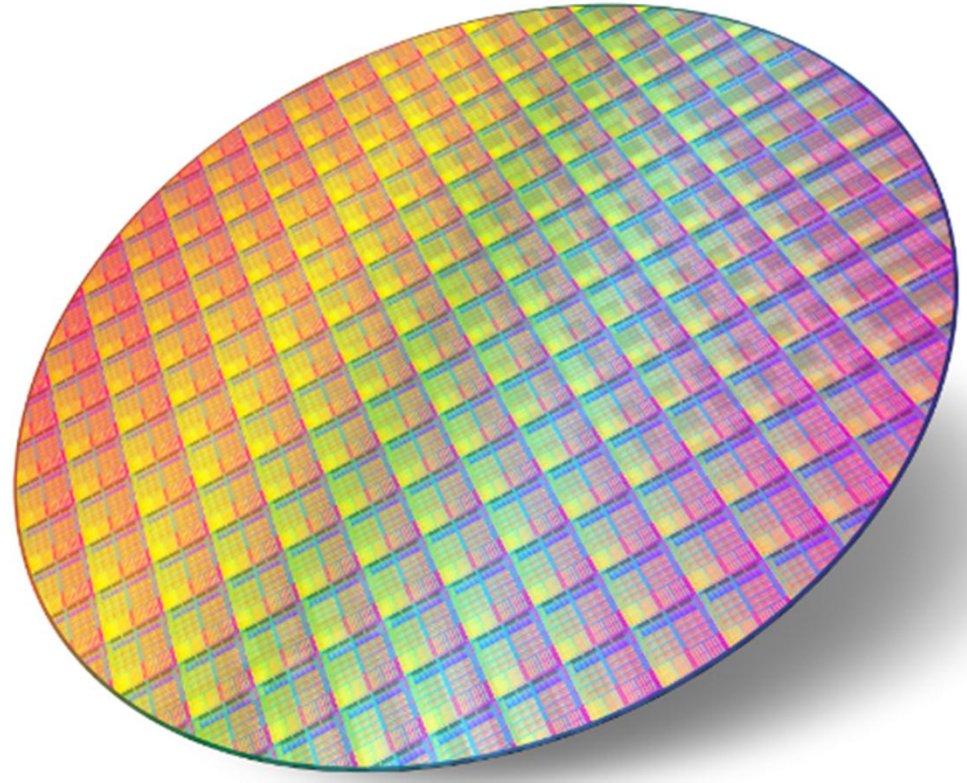
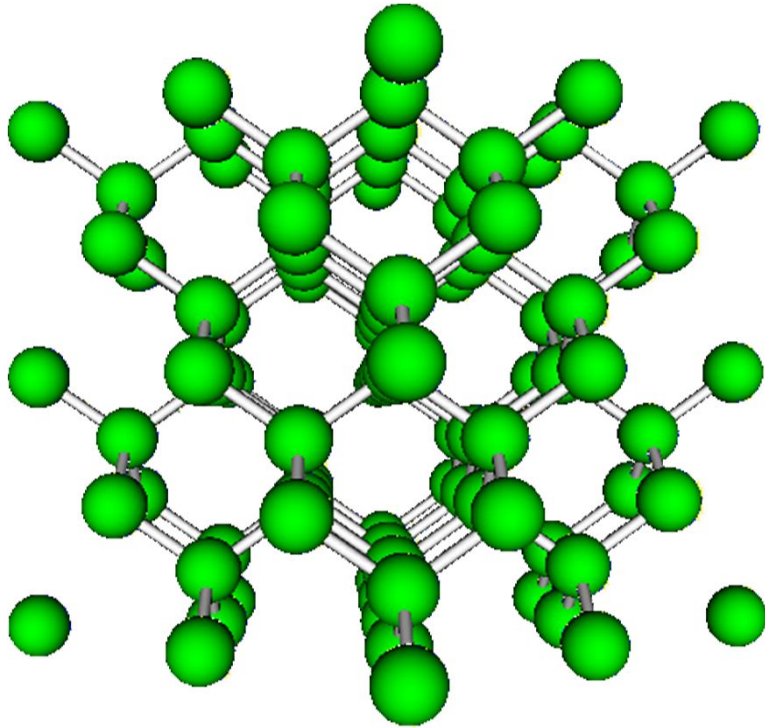
Gate for MOSFETs

Semiconductor layer
for TFTs (recent
development)



Polycrystalline silicon has higher conductivity as compared to amorphous silicon.

Crystalline Si



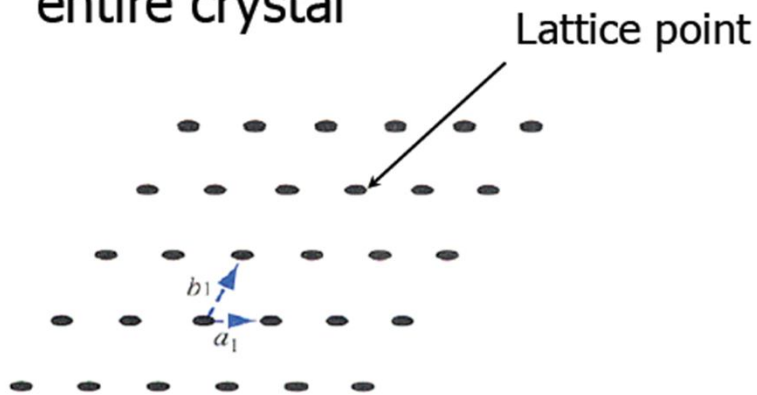
Very high performance and is used to make ICs

Lattice and Unit Cell

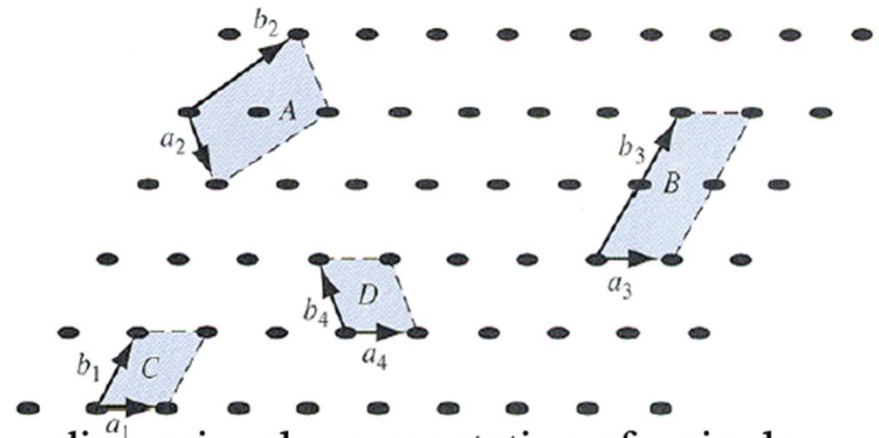
Lattice : A regular periodic arrangement of points in space as in the arrangement of atoms or molecules in a crystal.

Each point in the lattice is a lattice point. It can be an atom, a group of atoms, an ion or a molecule.

Unit cell : A small volume of a crystal that can be used to reproduce the entire crystal



Two dimensional representation of a single crystal lattice

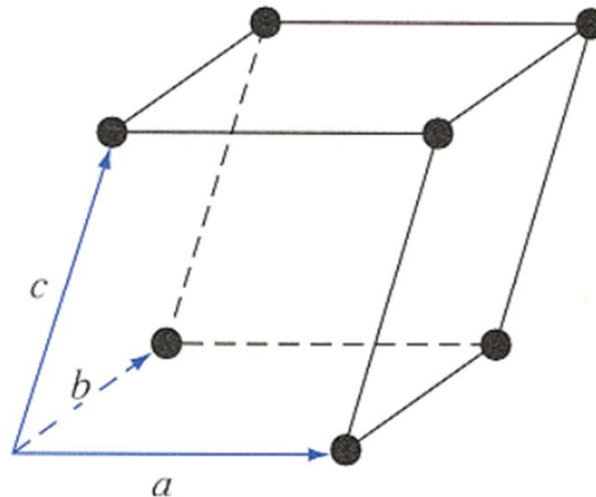


Two dimensional representation of a single crystal lattice with various possible unit cells

Lattice, Unit Cell and Primitive Cell

The relationship between the cell and the lattice is characterized by three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$. These vectors need not to be perpendicular to each other and need not to be of the same length. Every equivalent lattice point in the crystal can be found using the equation –

$$\vec{r} = p\vec{a} + q\vec{b} + s\vec{c} \quad (1.1)$$



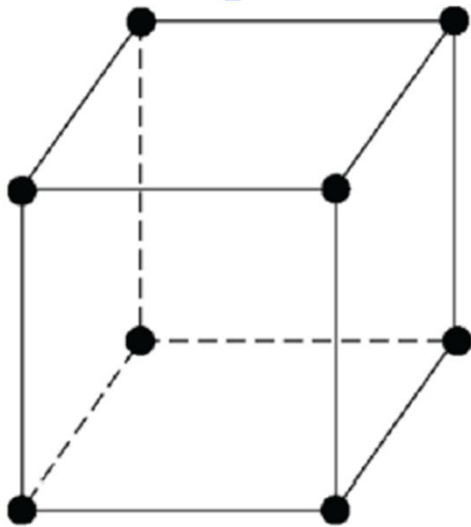
A generalized primitive unit cell

A **PRIMITIVE CELL** is the *Smallest* unit cell!

Basic Crystal Structures

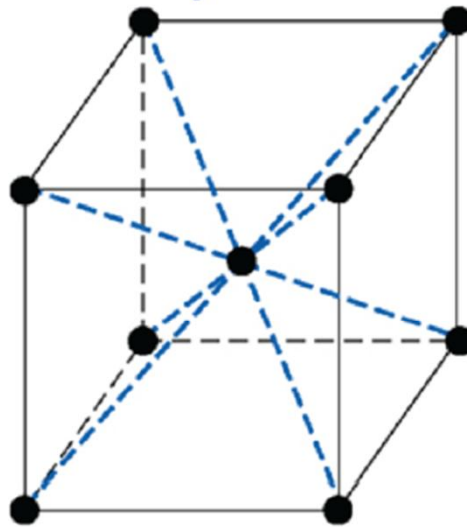
The easiest 3-D lattice to work with is the simple cubic lattice (SCC) which has lattice points on all the corners of a cube. The Cubic (Isometric) crystal system is characterized by its total symmetry. It has three crystallographic axes that are all perpendicular to each other and equal in length. The cubic system has one lattice point on each of the cube's eight corners.

Simple



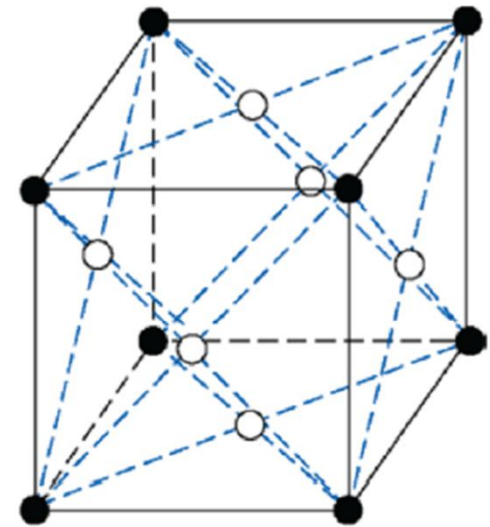
(a)

Body centered



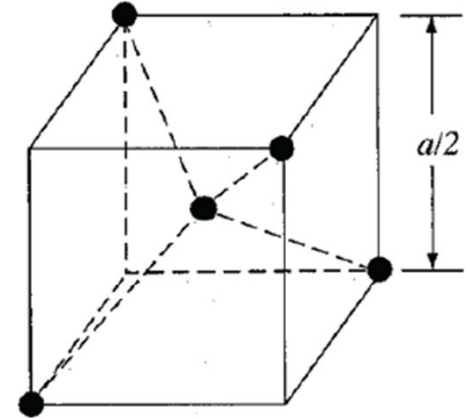
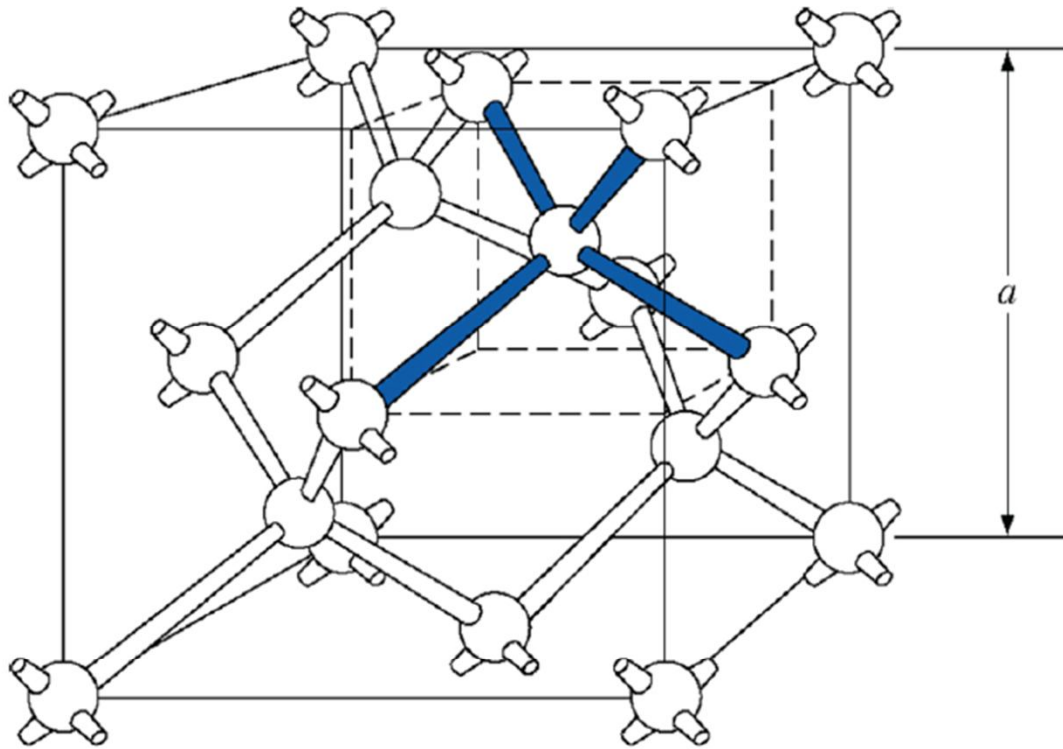
(b)

Face centered



(c)

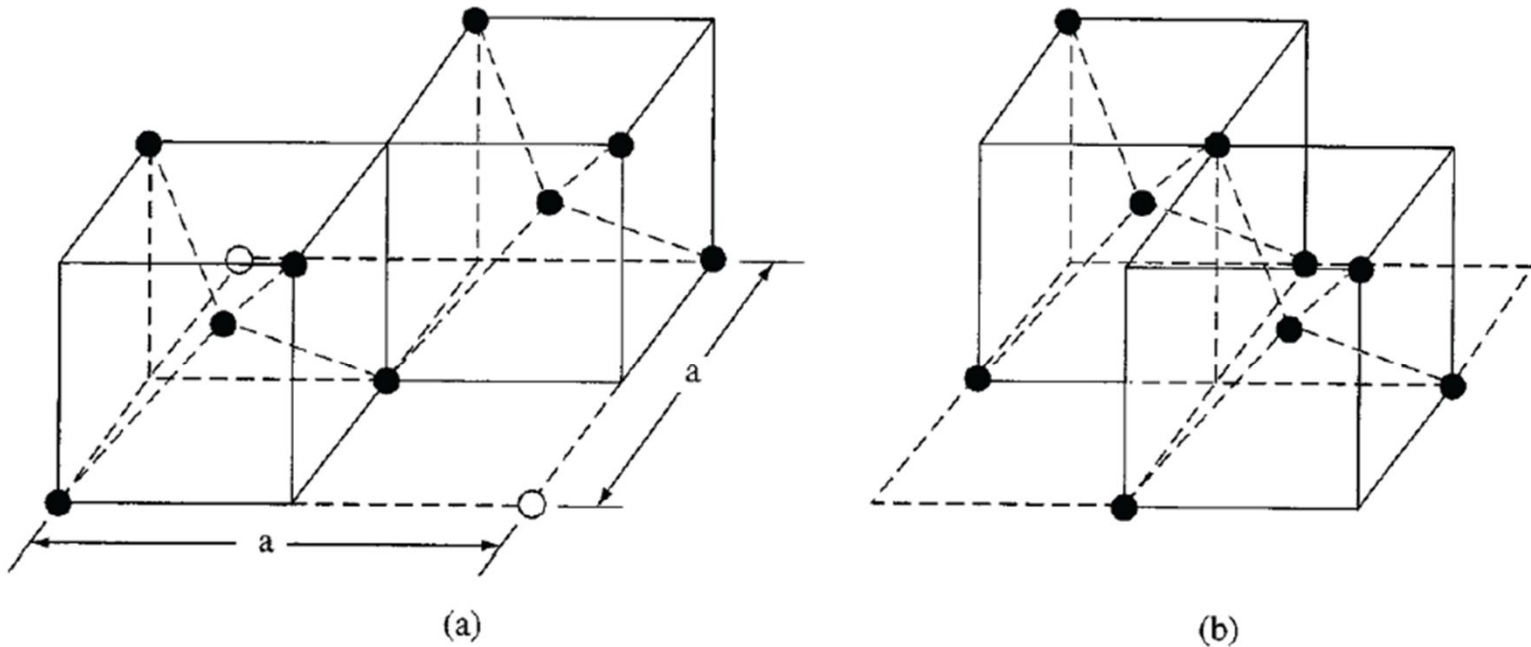
The Diamond Structure (Si/Ge)



The tetrahedral structure of closest neighbors in the diamond lattice

The tetrahedral structure is basically a body-centered cubic with four of the corner atoms missing. Every atom in the tetrahedral structure has four nearest neighbors with equal distance. This is the basic building block of the diamond lattice.

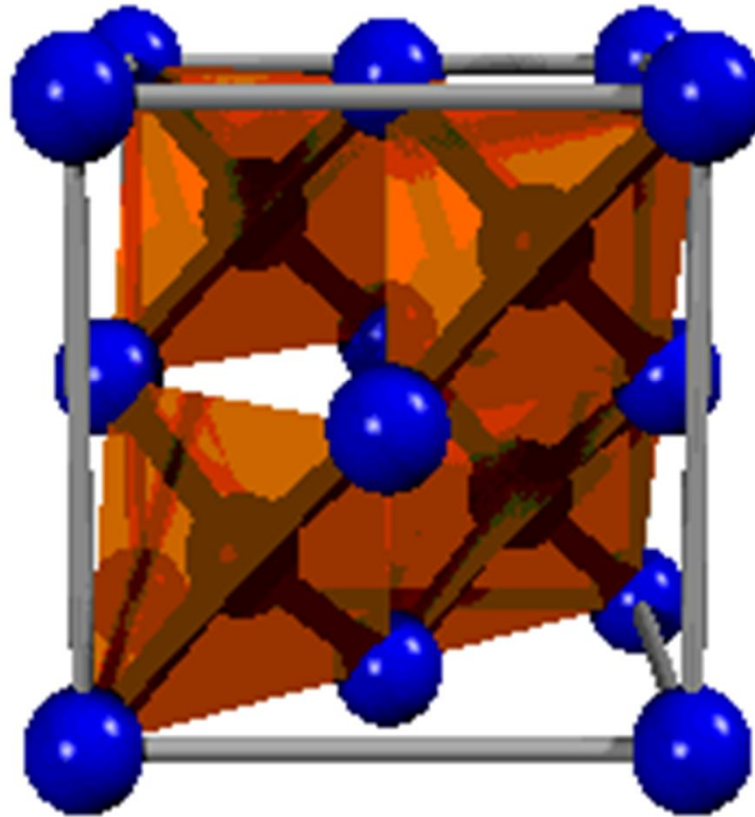
The Diamond Structure



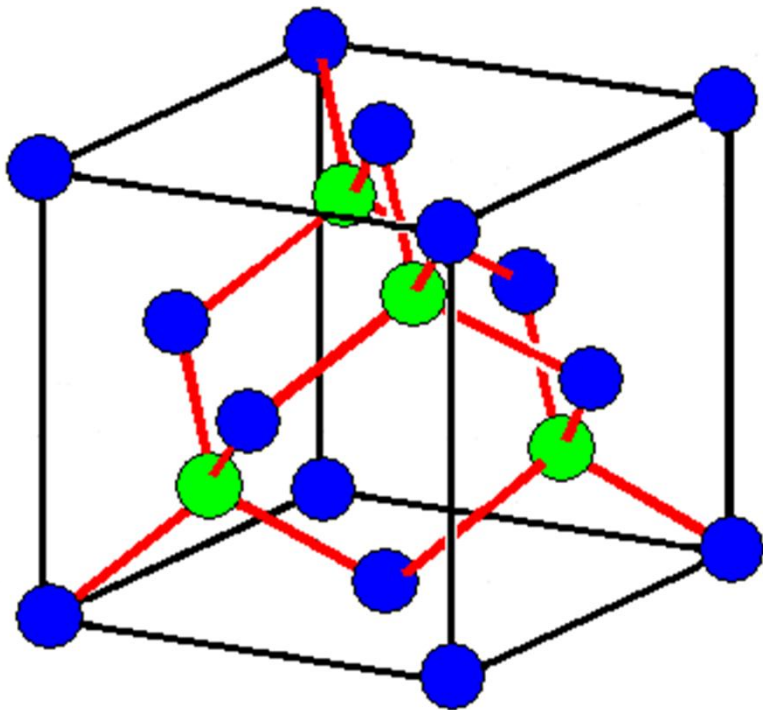
Bottom half (a) and top half (b) portions of diamond lattice.

- Silicon and Germanium have this structure
- All atoms are of the same species
- They are bonded by covalent bonding

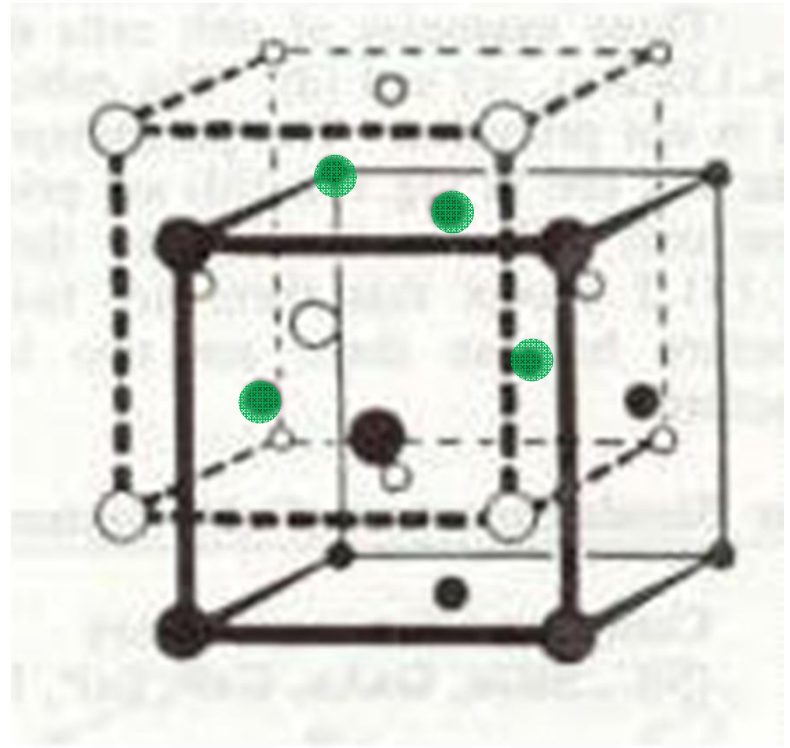
The Diamond Structure



The Diamond Structure

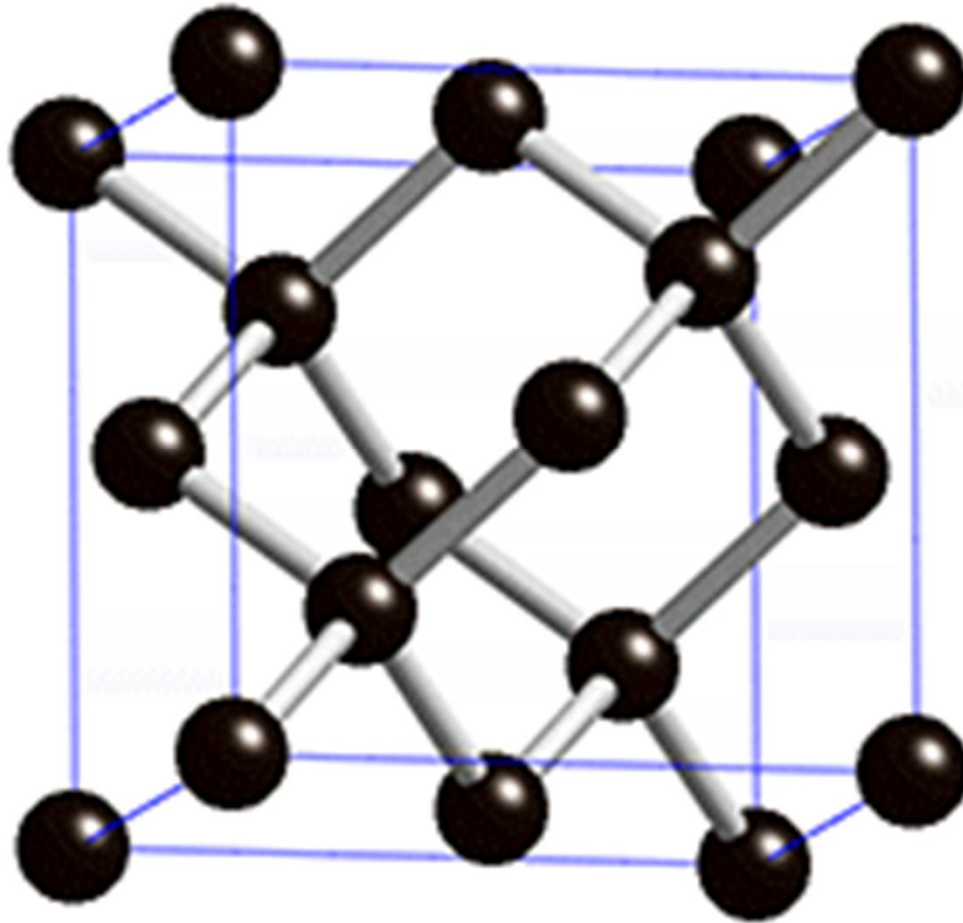


- Blue: atoms of a fcc cell
- Green: new atoms introduced to a fcc cell
- Diamond unit cell = fcc cell + 4 new atoms

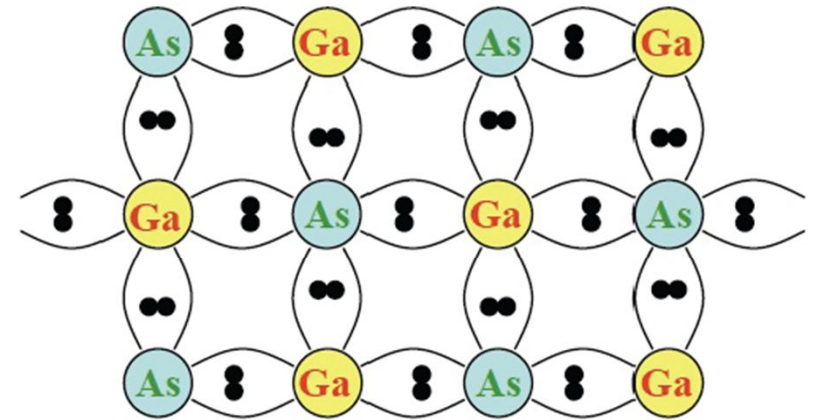
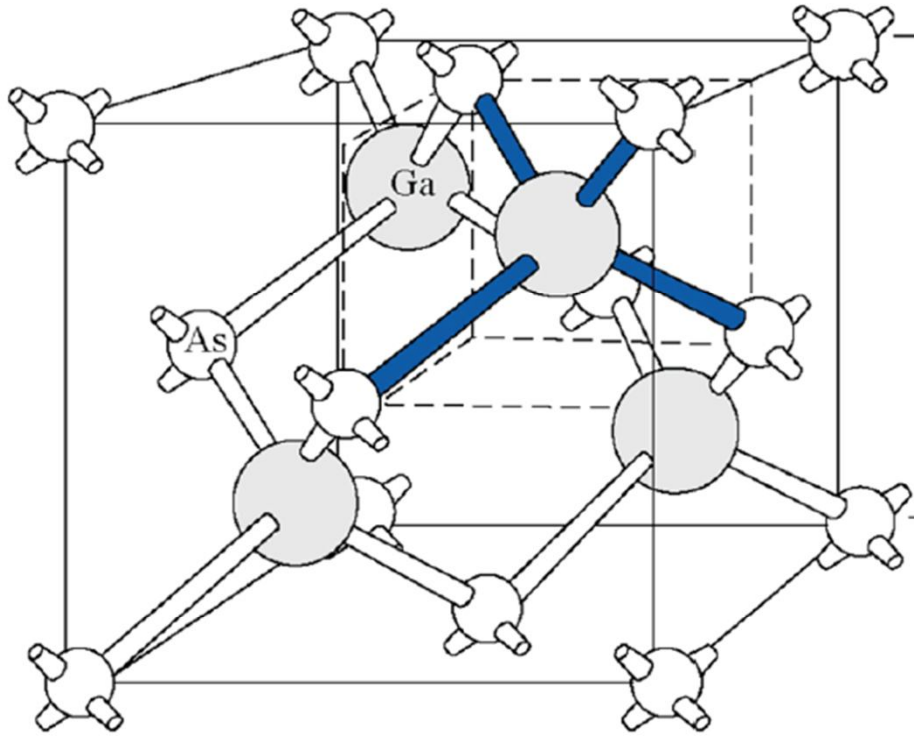


The space lattice of diamond is fcc. It is composed of two fcc lattices displaced from each other by $\frac{1}{4}$ of a body diagonal, $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

The Diamond Structure



The Zincblend Structure (GaAs)



$_{31}\text{Ga}:(\text{Ar})3d^{10}4s^24p^1$

$_{33}\text{As}:(\text{Ar})3d^{10}4s^24p^3$

Covalent + Ionic bonding

Zincblend structure is similar to diamond structure, but atoms in penetrating FCC are different.

Zincblend 闪锌矿

Crystal plane and Miller Indices

Surface or plane through the crystal can be described by the intercepts of the plane along the $\vec{a}$, $\vec{b}$ and $\vec{c}$ axes used for the lattice. Reciprocal of these intercepts are called the miller indices.

Intersect ions:

3, 2, 1

Reciprocals:

$1/3, 1/2, 1/1$

Multiply by 3x2x1:

2, 3, 6

Miller plane indices:

(2 3 6)

General plane:

(h k l)

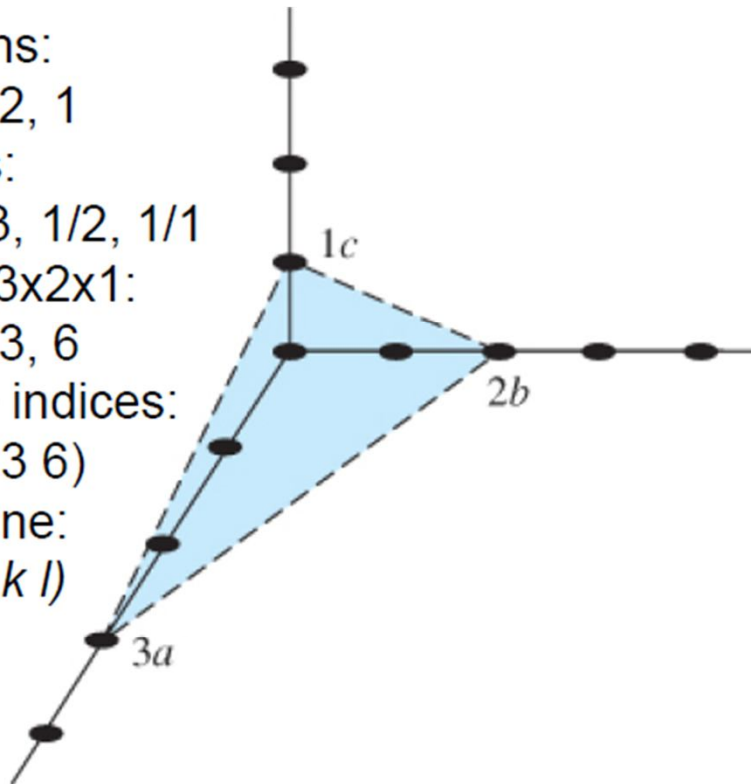


Figure 1.6 | A representative crystal-lattice plane.

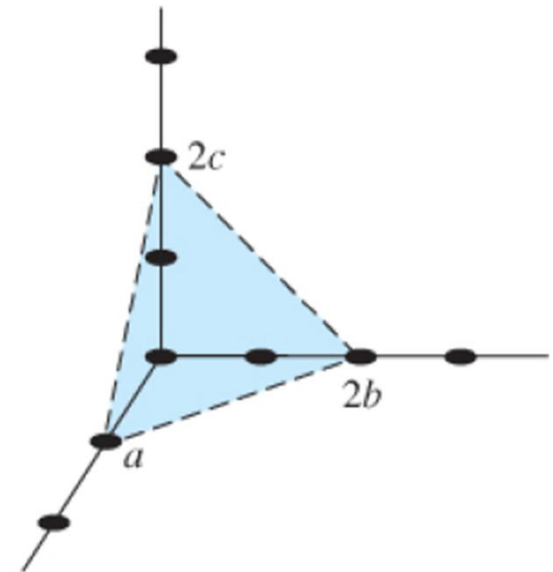
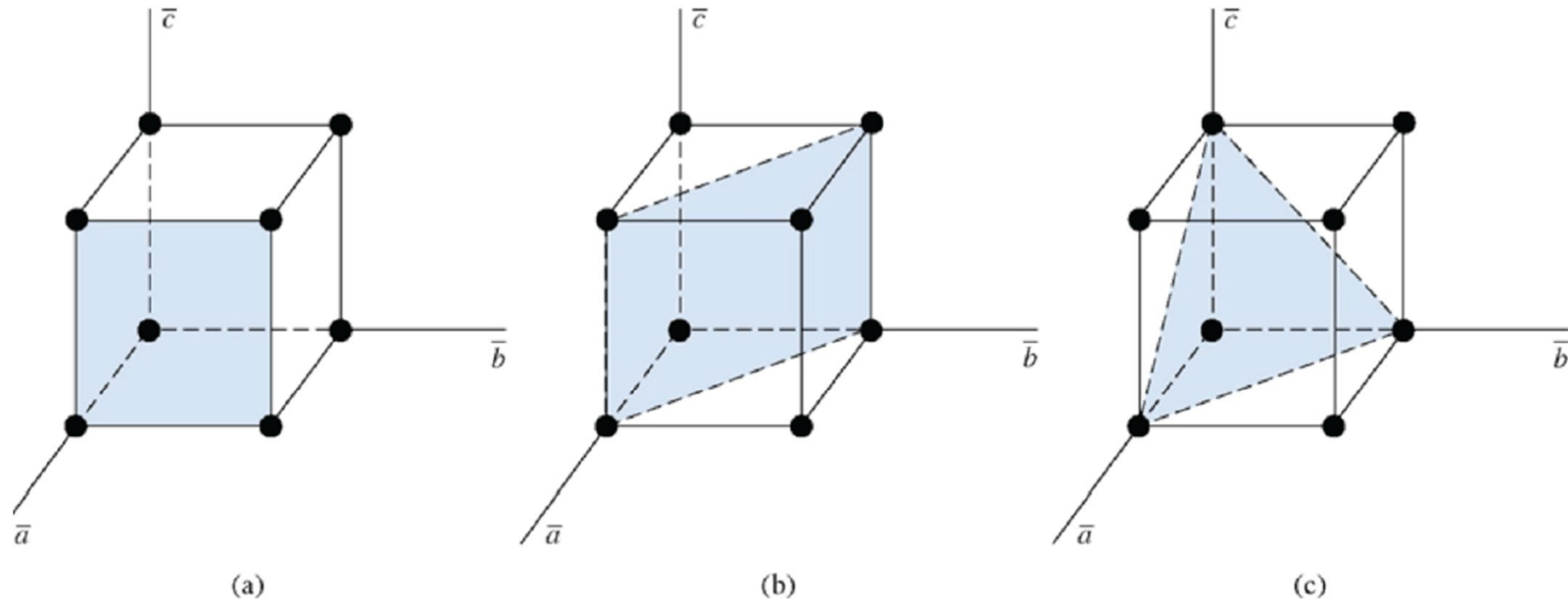


Figure 1.7 | Figure for Exercise Problem Ex 1.2.

$$1, 2, 2 \rightarrow 1, \frac{1}{2}, \frac{1}{2} \rightarrow (2 \ 1 \ 1)$$

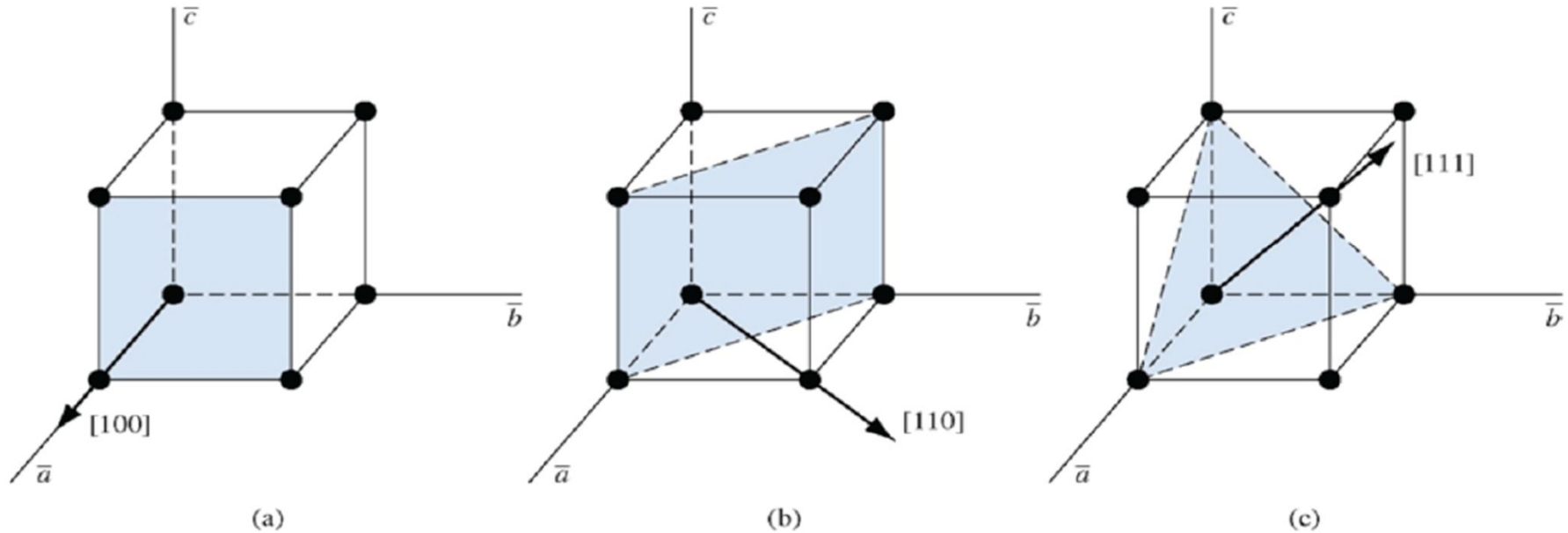
Crystal plane and Miller Indices

Surface or plane through the crystal can be described by the intercepts of the plane along the $\vec{a}$, $\vec{b}$ and $\vec{c}$ axes used for the lattice. Reciprocal of these intercepts are called the miller indices.



Three lattice planes – (a) (100) plane, (b) (110) plane and (c) (111) plane

Directions in Crystals

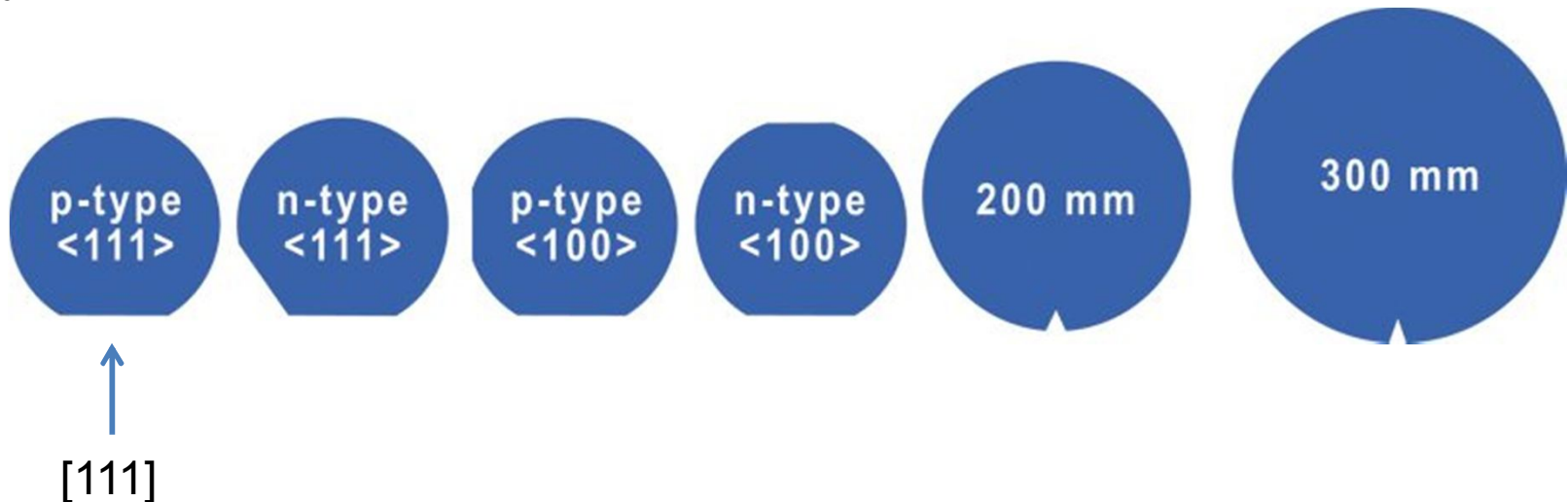


The plane (a) is parallel to the $\vec{b}$ and $\vec{c}$ axes so the intercepts are given as 1, infinite and infinite. Taking reciprocal, we obtain Miller indices as (1,0,0), so the plane is referred to as (100) plane. Any plane parallel to this is equivalent and is referred to as the (100) plane.

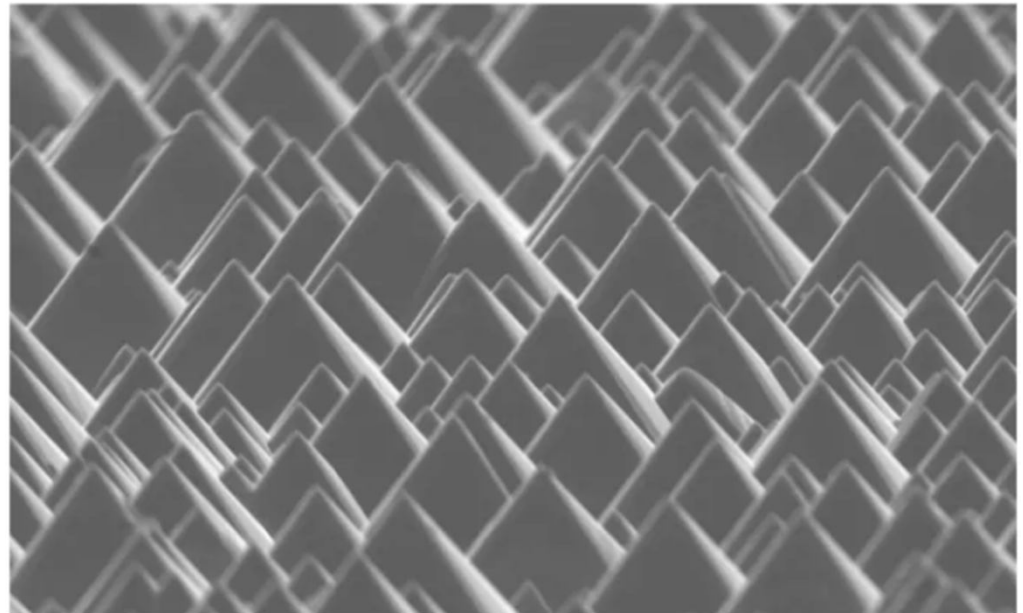
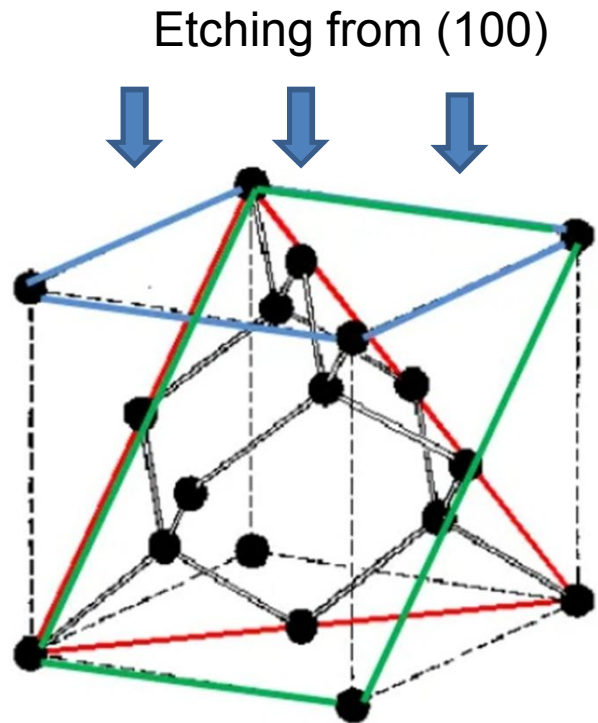
Three lattice directions and planes: (a) (100) plane and $[100]$ direction, (b) (110) plane and $[110]$ direction, (c) (111) plane and $[111]$ direction.

Silicon Wafer Flats to Determine Type/Doping and Orientation

- Orientation flats that indicate the silicon doping and orientation for wafers of less than 200 mm diameter.
- For wafers with diameters less than 200 mm, the primary (largest) flat is oriented perpendicular to a specified crystal axis such as $\langle 111 \rangle$ or $\langle 100 \rangle$.
- Secondary (smaller) flats indicate whether a wafer is either p-type or n-type.
- 200 mm (8-inch) and 300 mm (12-inch) wafers use a single notch oriented to the specified crystal axis to indicate wafer orientation with no indicator for doping type.



Crystal plane and etching anisotropy



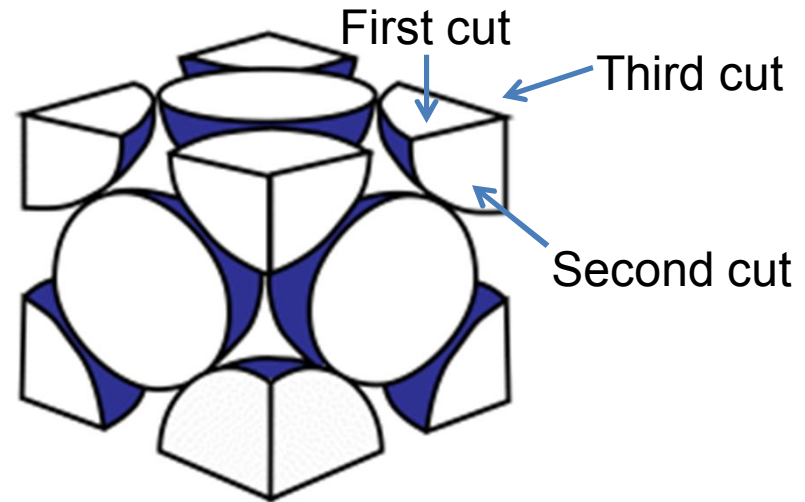
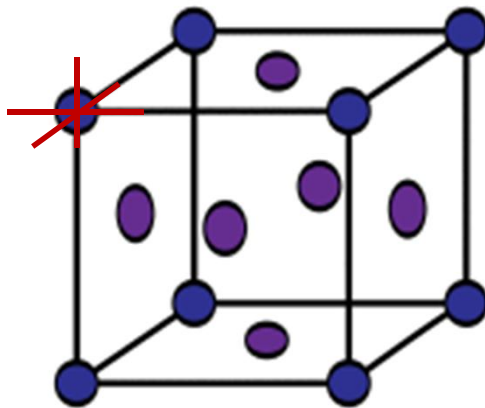
金字塔的侧面由(111)面构成，地面由(100)面构成

- 不同晶面的硅原子间具有不同的能量，碱液刻蚀过程呈现出各向异性
- Etching speed: $(100) > (111)$
- In Si, the (111) plane has a closely-packed arrangement of atoms, which leads to a relatively low surface energy. The (100) plane, on the other hand, has a higher surface energy due to the presence of broken bonds and dangling bonds on the surface. The higher surface energy of the (100) plane makes it more reactive to chemical etchants, as the bonds on the surface can be more easily broken.

Volume density and Surface density

By knowing the crystal structure and the lattice dimensions, we are able to determine the surface density (# atoms/cm²) and volume density of atoms (# atoms/cm³)

Two representations of FCC Crystal Structure



From the right figure above we see that there is $1/8$ of an atom at each of 8 corners of the unit cell and $1/2$ atom at each of 6 faces. This gives $(8 \times 1/8) + (6 \times 1/2) = 4$ atoms per crystal cell.

Example (1): volume density of Si

How Many Silicon Atoms per cm^{-3} ?

- **Number of atoms in a unit cell:**

- 4 atoms completely inside cell
- Each of the 8 atoms on corners are shared among cells
→ count as 1 atom inside cell
- Each of the 6 atoms on the faces are shared among 2
cells → count as 3 atoms inside cell

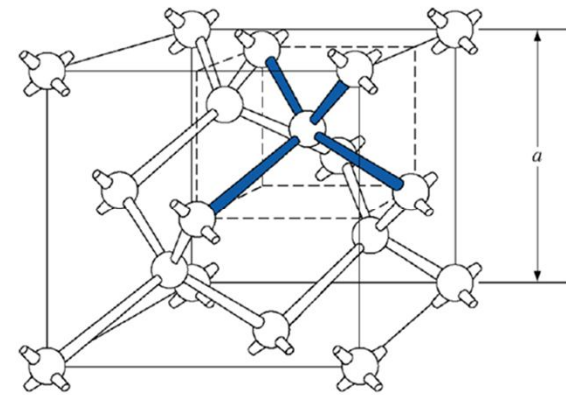
⇒ Total number inside the cell = $4 + 1 + 3 = 8$

- **Cell volume:**

$$(.543 \text{ nm})^3 = 1.6 \times 10^{-22} \text{ cm}^3$$

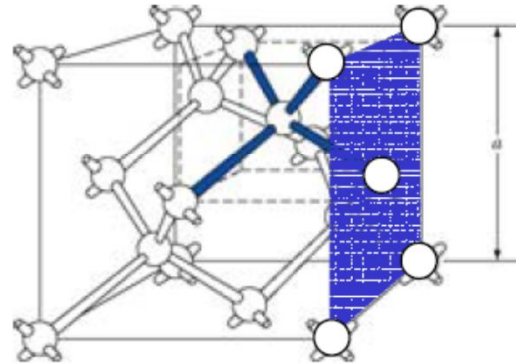
- **Density of silicon atoms**

$$= (8 \text{ atoms}) / (\text{cell volume}) = 5 \times 10^{22} \text{ atoms/cm}^3$$

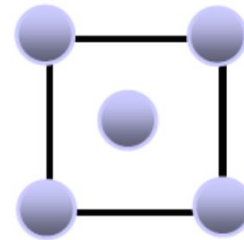


Example (2): surface density of Si

Calculate the surface density of atoms (# atoms/cm²) on the shaded plane (010 plane) of a silicon crystal shown. The lattice constant for silicon is 5.43 Å (1 Å = 10⁻¹⁰ m).



The silicon crystal has the face-centered cubic structure. The indicated plane can be redrawn as below



Example (2): surface density of Si

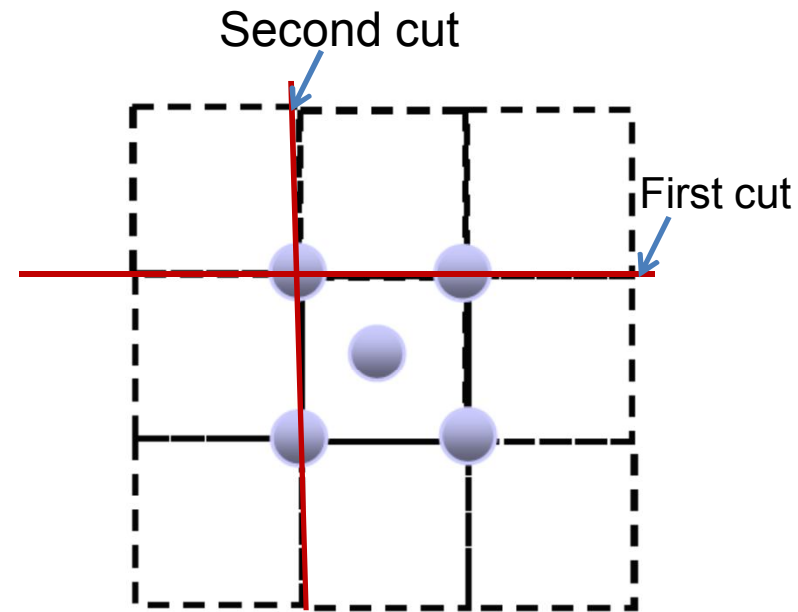
Notice that the atom at each corner is shared by 4 similar equivalent lattice planes, so each corner atom effectively contributes $1/4$ atom to the indicated lattice plane.

The atom in the middle of the plane is not shared by any other plane.

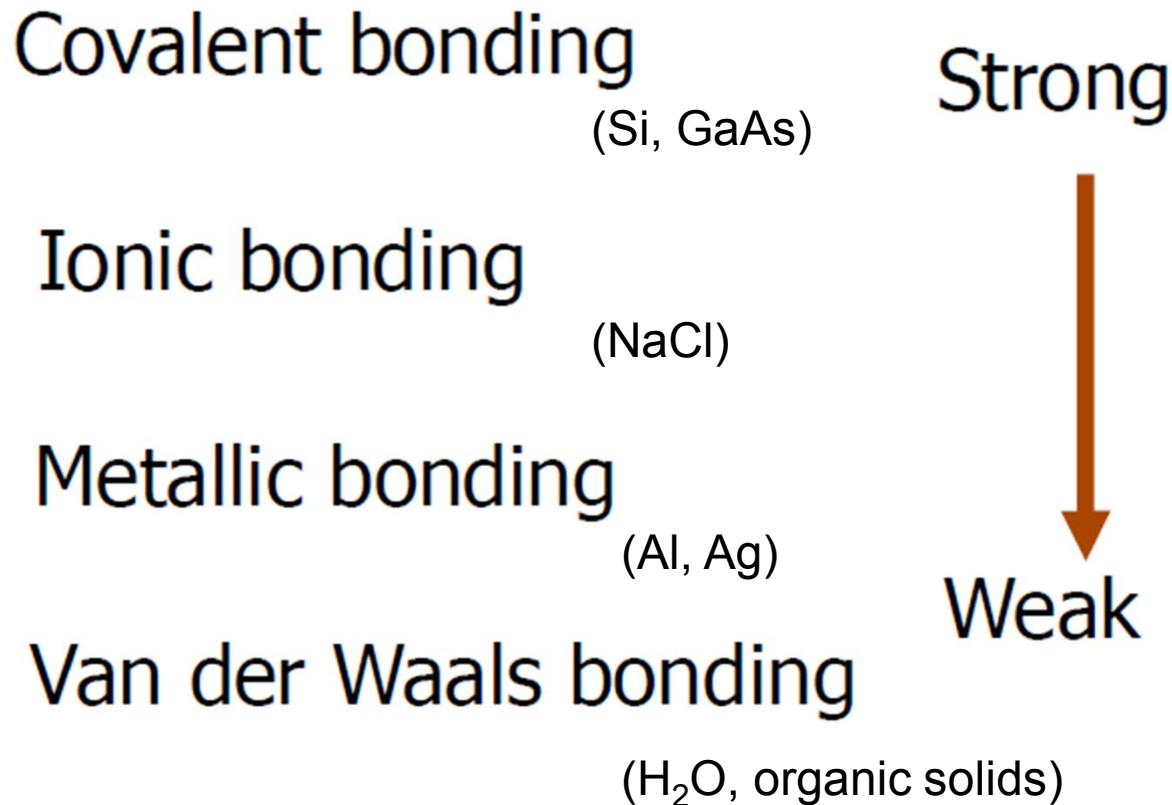
Hence, no. of atoms on the plane is $4(1/4) + 1 = 2$ atoms

Therefore,

$$\begin{aligned}\text{Surface density} &= \frac{\text{No. of atoms}}{a^2} = \frac{2}{(5.43 \times 10^{-8} \text{ cm})^2} \\ &= 6.78 \times 10^{14} \text{ atoms/cm}^2\end{aligned}$$

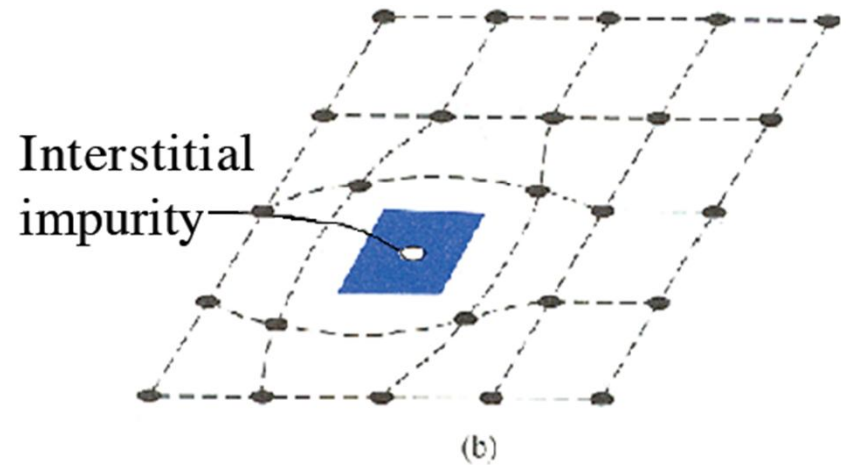
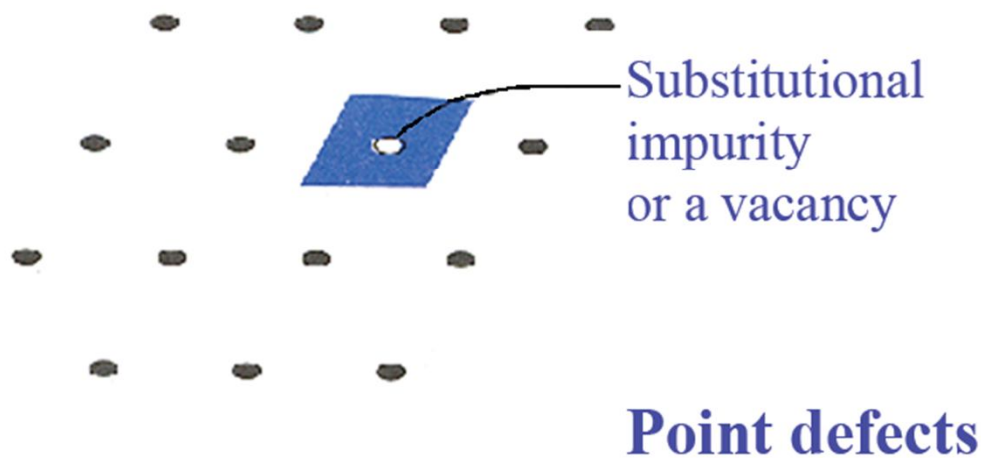


Atomic Bonding

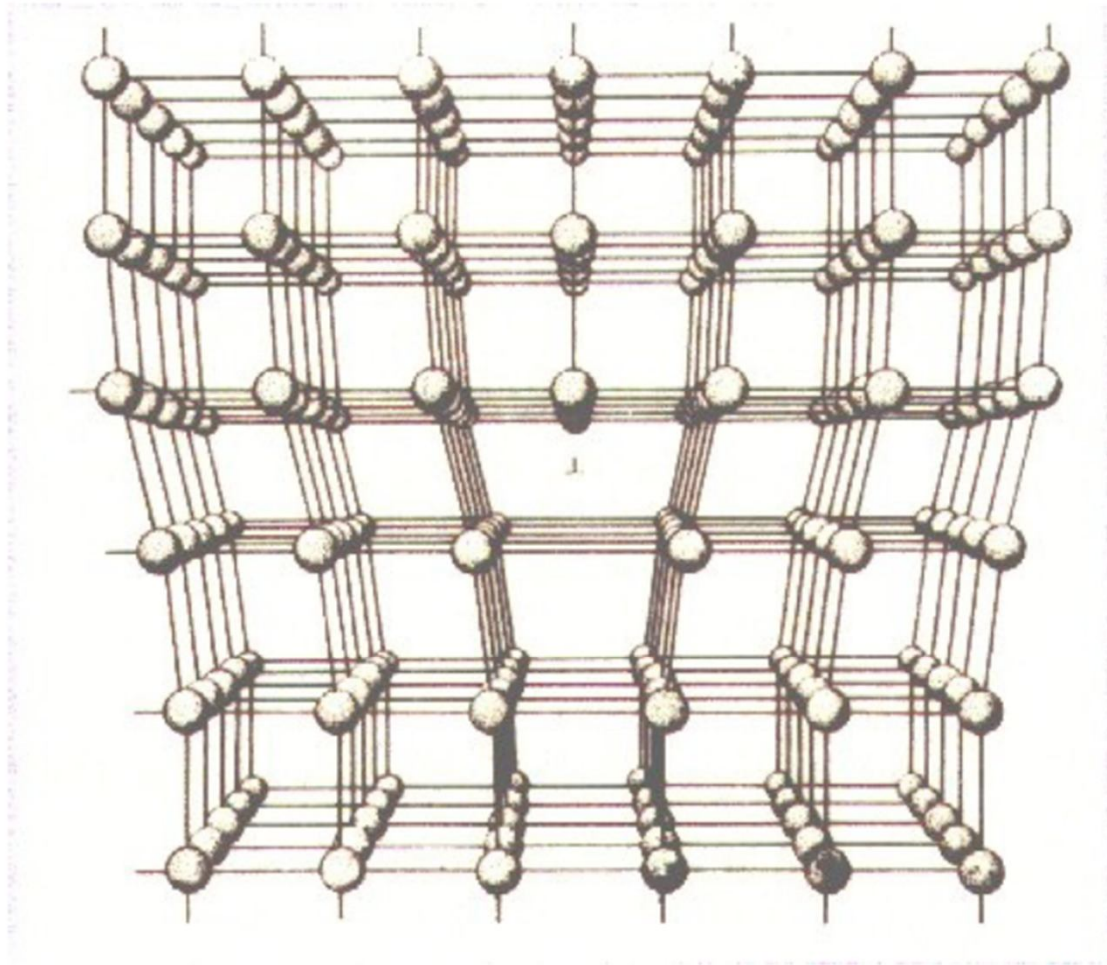


Imperfections in Solids

Lattice imperfection is any deviation from perfect periodic arrangement of lattice points. Point defects can be in the form of vacancy (missing atom), substitutional impurity (atom replaced by another atom) or interstitial impurity (additional impurity not at lattice point)

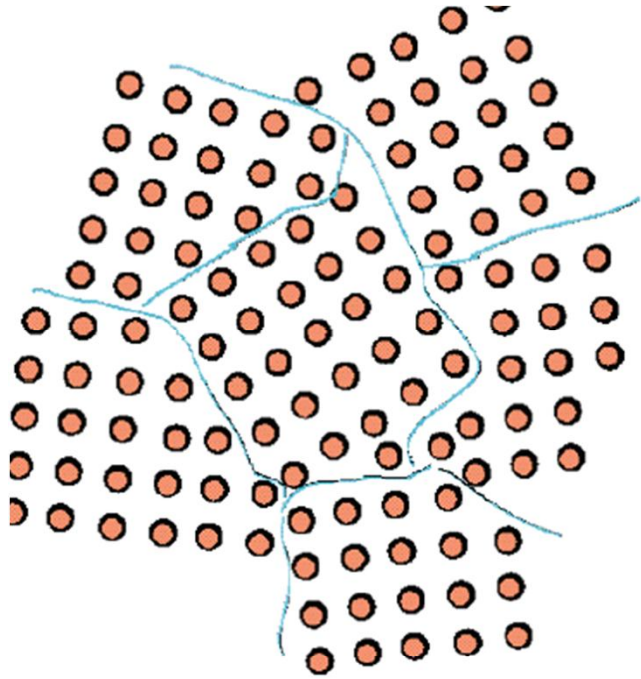


Line Defects

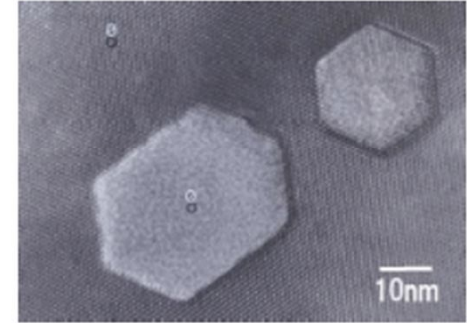


Edge dislocation is a type of line defect.

Plane & Volume Defects



Grain boundaries are
type of plane defect



Precipitates are a type
of volume defect

Production steps of silicon

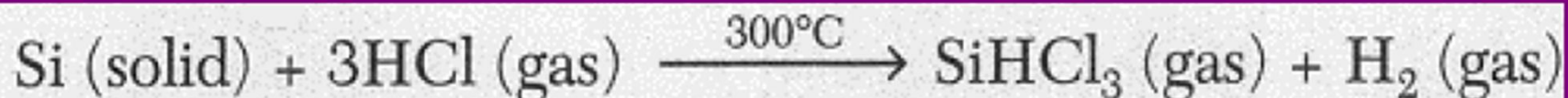
“95% of materials used by the electronic industry is silicon”

SiO₂ sand (quartzite)

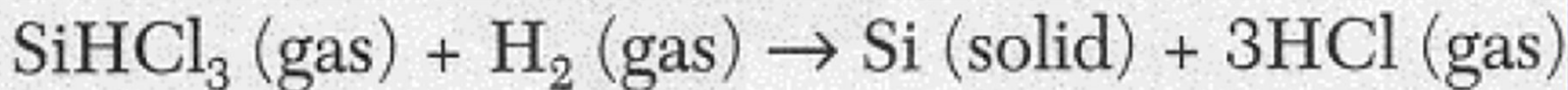
2000°C



98% pure silicon



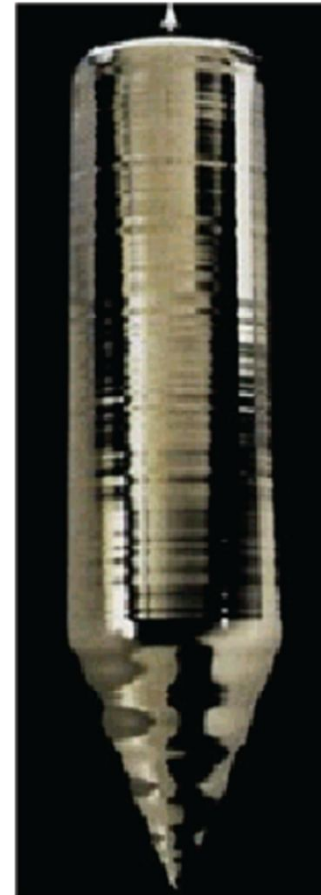
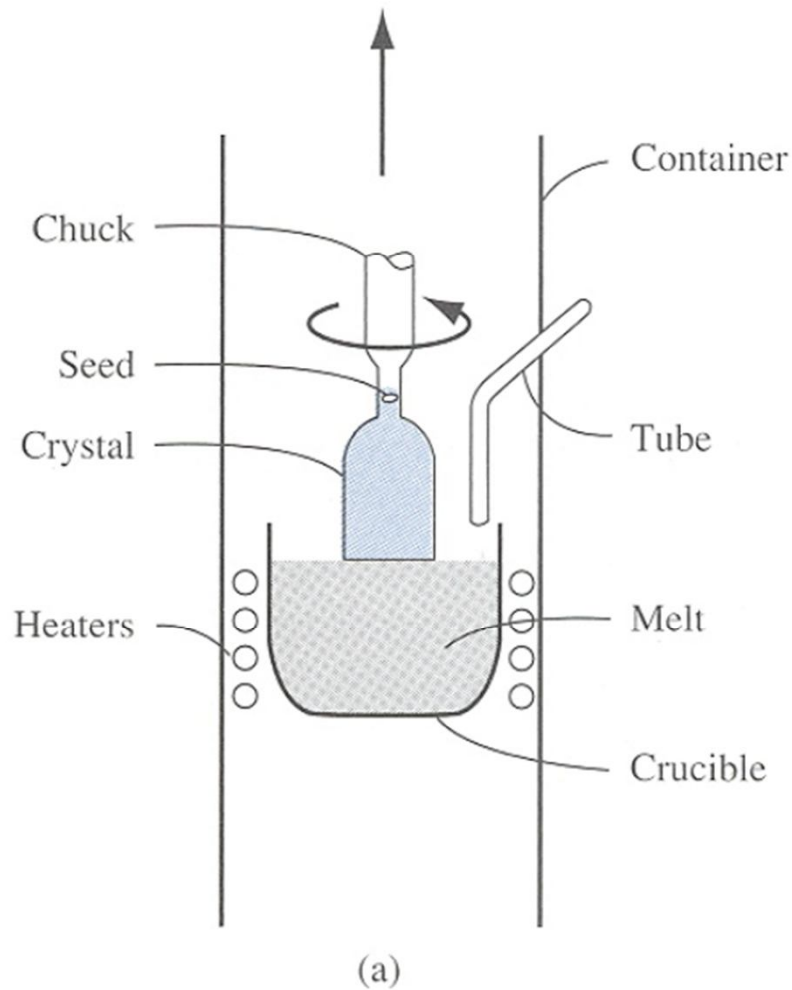
gaseous trichlorosilane



rods of ultrapure polycrystalline silicon

Growth of Semiconductor Materials

切克劳斯基提拉法 Czochralski method



Silicon ingot
from
czochralski
process

Larger diameter , impurity from
crucible

Raw material of Silicon
(pure sand SiO_2)



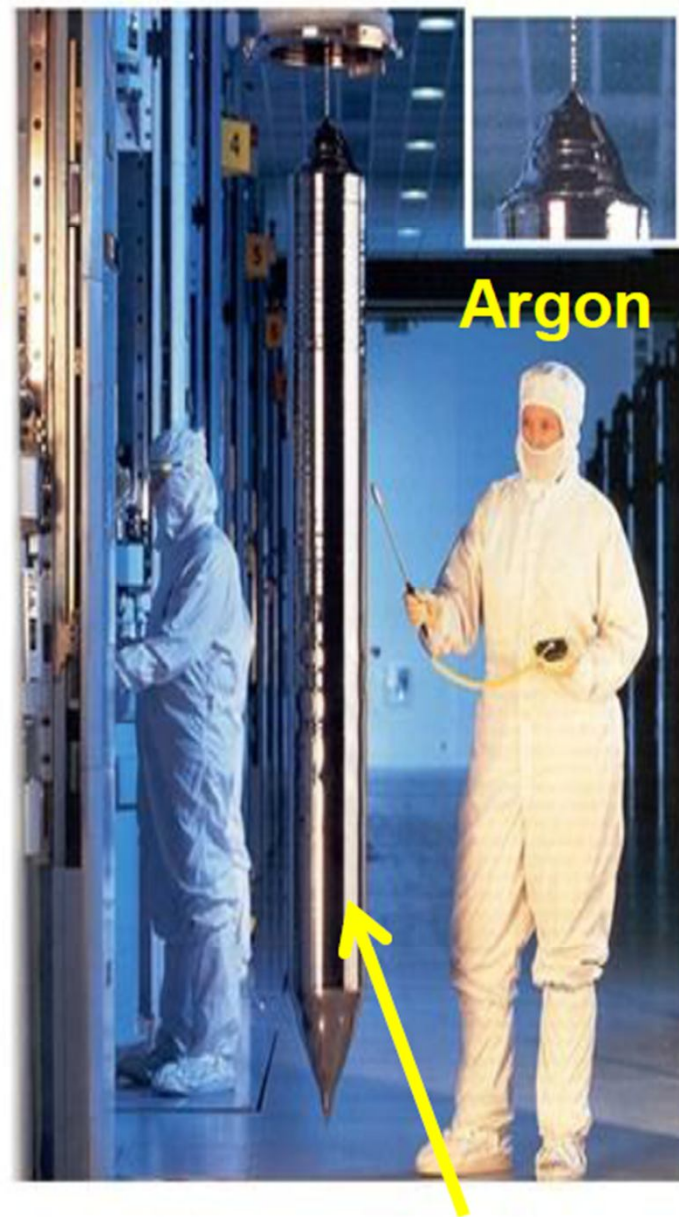
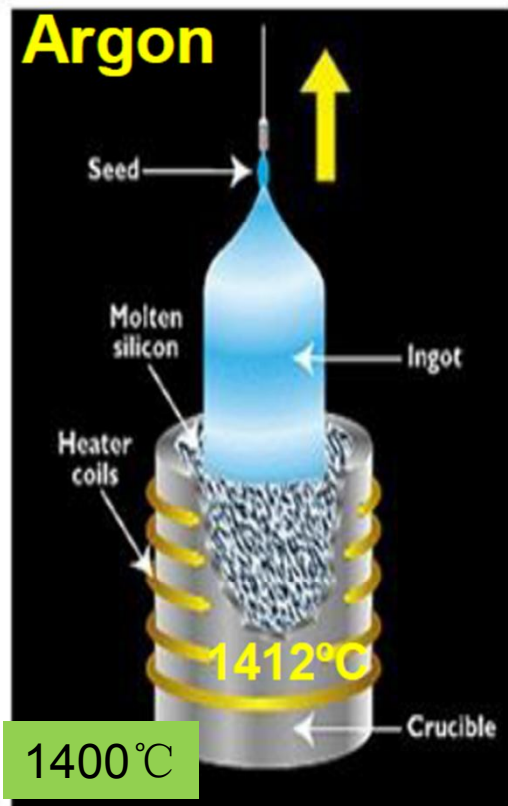
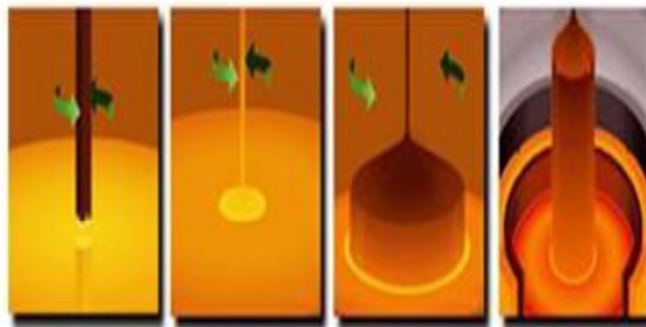
Step 1: Preparing the Molten Silicon Bath

The sand (SiO_2) is put into a rotating crucible and is heated above its melting point. The molten sand will become the source of the silicon that will form the wafer.

Step 2: Making the Ingot

A pure silicon seed crystal is now placed into the molten sand bath. This crystal will be pulled out slowly as it is rotated.

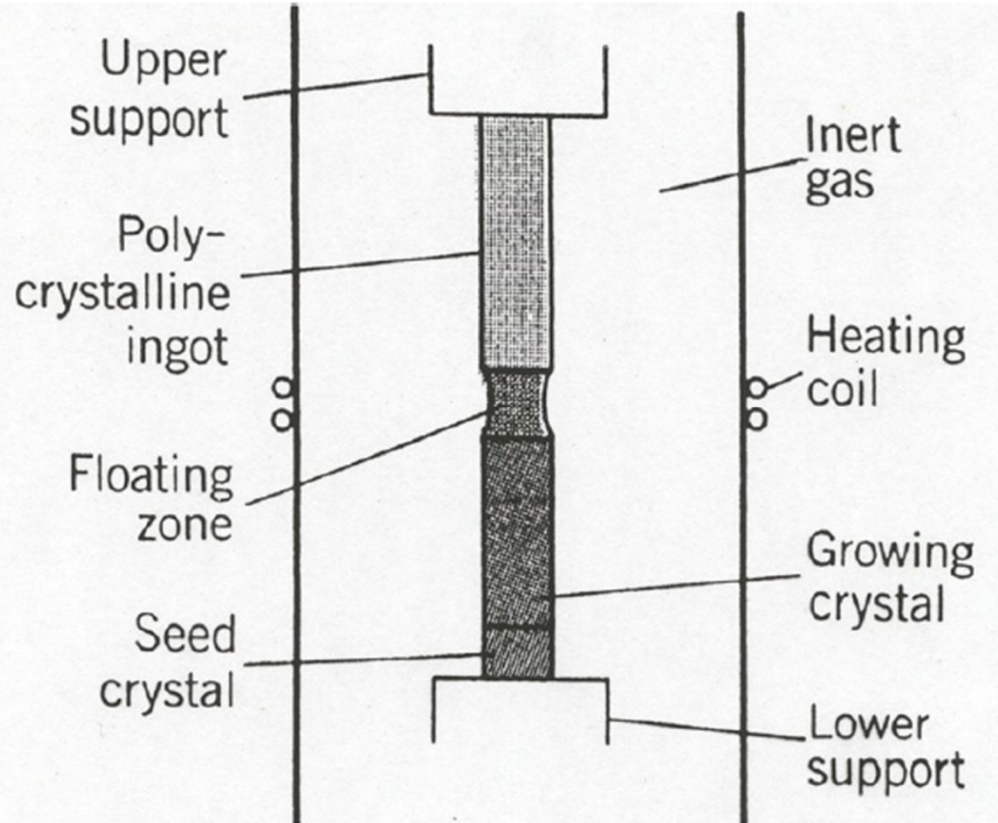
Czochralski technique



Argon

Silicon ingot

Floating zone method for purifying silicon



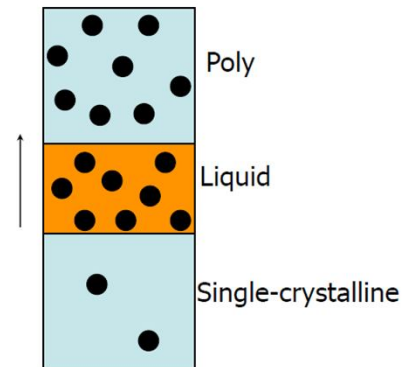
浮区法

Floating zone method relies on the segregation co-efficient. For this process to be used for purification this co-efficient should be less than 1.

Segregation coefficient (k_0)

Impurity concentration ratio between the solid and liquid

$$k_0 = \frac{C_{Solid}}{C_{Liquid}}$$



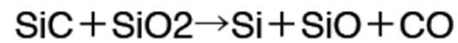
	k_0	$D \times 10^{-4} \text{ cm}^2/\text{s}$
B	0.8	2.4
Al	0.002	7.0
Ga	0.008	4.8
In	0.0004	6.9
C	0.07	2
P	0.35	2.3
As	0.3	3.3
Sb	0.023	1.5
O	1.40	3.3

Si wafer fabrication

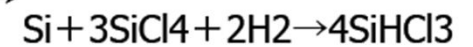
SiO₂



Metallic Si



Pure trichlorosilane

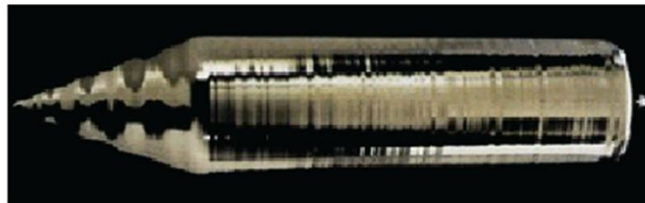


Pure poly-Si

99.999999999%



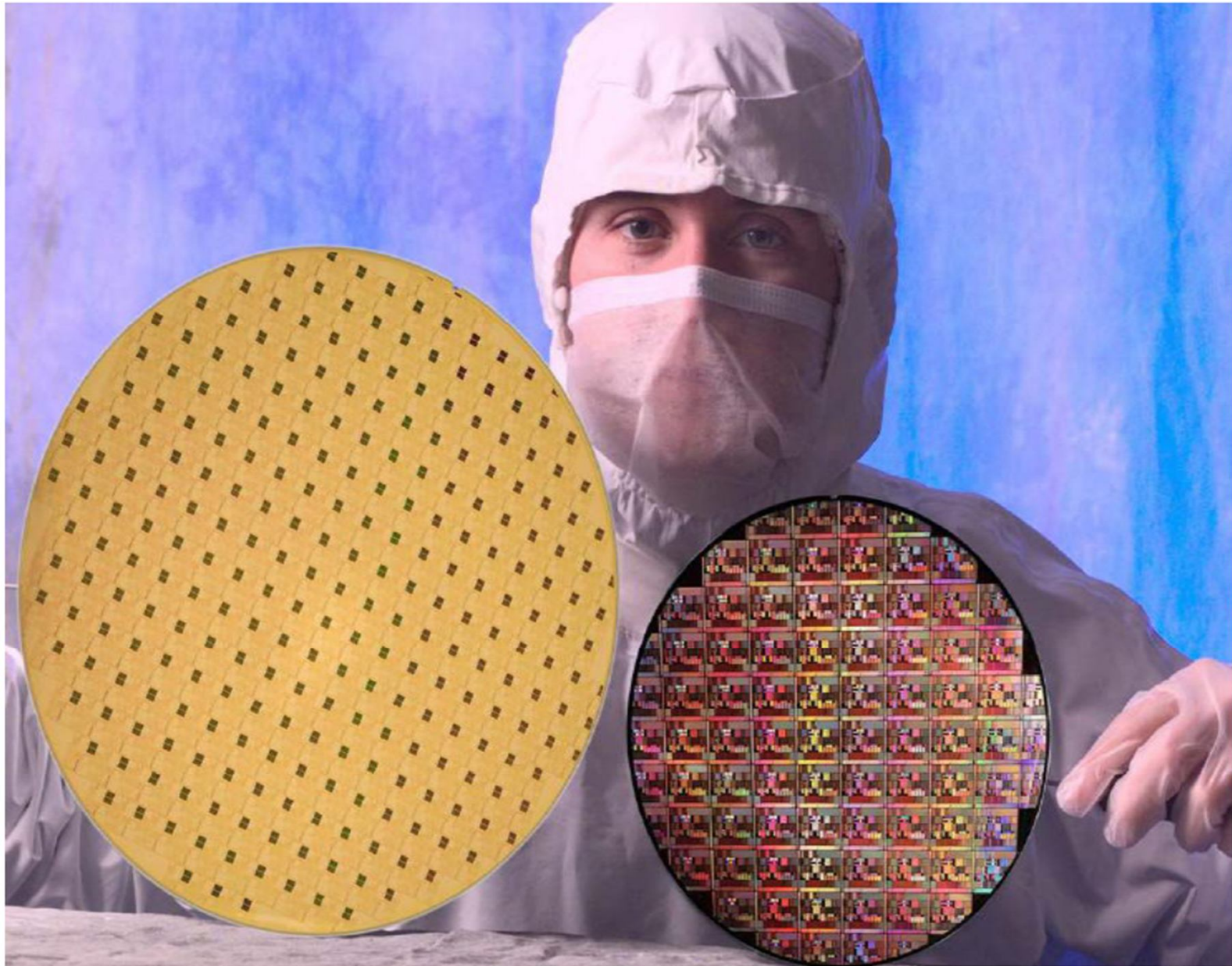
Single-crystalline Si



Wafer



8 inch (200 mm) and 12 inch (300 mm) wafer



Epitaxial growth

Single-crystalline layer growth on a single-crystalline substrate with the same crystal structure

Homo-epitaxy

Si on Si

Hetero-epitaxy

Si on SiGe, AlGaAs on GaAs

by means of

Chemical vapor deposition (CVD)

Liquid Phase Epitaxy (LPE)

Molecular Beam Epitaxy (MBE)

Reading materials

D. A. Neamen, “Semiconductor Physics and Devices: Basic Principles”, Fourth edition, Chapter 1, “the Crystal Structure of Solids”.